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1 Introduction

Ragwort (*Jacobaea vulgaris*) is a widespread field weed whose presence in grazing areas poses a direct threat to the health and life of livestock. Due to its high toxicity, this plant requires strict and systematic elimination from pastures. In the era of increasing demand for the optimization and automation of agricultural processes-commonly referred to as precision agriculture-utilizing modern computer vision methods for the detection of harmful flora species has become a key research and engineering challenge.

2 Problem Statement

The current standard for identifying and removing ragwort relies on manual field inspection. This process is highly inefficient, lacks scalability, and is extremely time-consuming. Furthermore, this plant features a high degree of visual camouflage among natural, tall pasture vegetation, which drastically reduces the effectiveness of the human eye in its early detection.

The primary technological barrier hindering the automation of this process is the lack of open, high-quality datasets. Most photographs available in public domains are macro shots that do not reflect the perspective of agricultural machinery (such as drones or rovers), making it impossible to effectively train classical vision models for environments encountered in the real world.

3 Project Goals and Proposed Solution

The main objective of this project is the design, implementation, and evaluation of an automated computer vision pipeline capable of precise, real-time ragwort detection.

To address the defined challenges, the project will achieve the following specific goals:

- **Dataset Development:** Creating, filtering, and optimizing a dedicated training dataset using image processing methods (including normalization, contrast enhancement, and the OpenCV library) alongside synthetic image generation.
- **Training and Evaluation of AI Models:** Parallel training and performance verification of two distinct deep learning architectures: YOLO family models (optimized for real-time processing speed) and models based on the DINO / DEIM architecture (focused on analyzing deep feature vectors, or embeddings, to maximize precision).
- **Visualization and Testing:** Building an analytical interface that enables the analysis of a live video stream while simultaneously generating bounding boxes around the detected plants.
- **Localization (Optional Goal):** Implementing a spatial mapping model (Facility Location) to provide the telemetry necessary for future autonomous systems designed for mechanical weed removal.

4 Proposed Work Pipeline

A basic Machine Learning Pipeline was adopted to achieve the aforementioned goals. The pipeline included steps mentioned in the succeeding sections with their own complexities being adapted to produce a better output upon evaluation.

Overall, the workflow went through four plans each adapted leading to the next one:

4.1 PLAN 1

At first, both the models were trained and tested with 1000 annotated images from our supervisor, Deepak Hanike. However, the dataset had its own limitations:

- It was too small to train the models effectively.
- The pictures were mostly biased close-up shots with poor quality.

- For YOLOv8 Nano with 3 million parameters, an accuracy of 65 % was achieved, which translated to negligible recognition in real-world.
- When tried on the DINOv3 model with 10 million parameters and 200 epochs, the result didn't meet an effective result either.

Hence, all the following plans adopted a larger open source dataset from GBIF, with variety of angles and distance shots and by far more images.

4.2 PLAN 2

The next plan was to try enhancing the images and filter out any undesirable images from the big dataset. However, since a significant change could not be achieved by enhancement, it was discarded as well.

4.3 PLAN 3

Plan 3 was substantially almost identical to our final workflow which included annotating a smaller mixed dataset selected from a larger set of images from GBIF. Then training our models with the labelled pictures to test them on the remaining 2800 images. This way the model would be then fine tuned and a higher accuracy would be achieved.

However, upon seeing the results it was seen that the model would recognize any yellow flower as ragwort and the mixed dataset didn't really help produce the desired effects.

4.4 PLAN 4

Finally, it was decided that a new labelled-top-view dataset would be prepared with the help of our supervisor. And simultaneously, a synthetic dataset placing ragworts in distant and different backgrounds would be prepared to fit the models to this new dataset and help it recognize the plant in more difficult-to-spot images.

5 Dataset Acquisition and Preprocessing

The training dataset was sourced from the Global Biodiversity Information Facility (GBIF) database. Occurrence records for ragwort observations were filtered to include only entries with associated images. Following retrieval of metadata and image references, a manual review process was conducted to assess suitability for model training. This evaluation ensured that selected images clearly depicted the target species and were representative of conditions expected in the final application environment.

Several image preprocessing techniques were explored using OpenCV to improve dataset quality. These techniques included:

- **Contrast enhancement** through histogram equalization
- **Image sharpening** via kernel-based filtering
- **Blur detection** using the variance of the Laplacian method

These approaches were evaluated to determine whether they improved visibility of relevant plant features. However, contrast enhancement and sharpening were **not included in the final preprocessing pipeline**, as they failed to provide significant benefits for the intended detection task.

Blur detection was tested as an automated quality control step but was not used in the tested method. A threshold was established using manually reviewed samples, enabling efficient filtering of large dataset portions while maintaining acceptable image quality. This approach did not substantially reduced the number of low-quality images requiring manual removal.

5.1 Sample Dataset Selection

After the desired bigger dataset would be prepared, a balanced diverse smaller sample dataset of around 2000 images would be selected using a pre-trained model called Facility Locations.

The initial choice was in favor of Facility Locations, since it chooses a balanced diverse dataset, which would not favor images of a certain kind. For instance: a majority of ragwort images with just flowers or just leaves. Also, the model usually removes blurred or defected images.

Since the nature of the project requires the robot to perform action on the plants and not only recognize them. It was established that even a mix of wide angle shots or images with complete plant visible from a side view would help the model learn to identify the plant even in midst of action, which might turn out to be helpful for future endeavors.

But after the chosen 200 images dataset was analysed, it was seen that the dataset had a big amount of bad images, which would lower the accuracy of our final models. Here, bad images included but was not limited to far away landscape shots with barely recognizable ragworts or distracting backgrounds such as animals, human hands, people, etc.

So, a change of method was proposed: The supervisor would select a new diverse dataset of 2000 images with minimal weight to noise using another algorithm. The selection would be performed from the synthetic dataset of 60000 images.

5.2 Annotations

To train the model from the sample dataset, annotations of the images would be needed.

Since, the models are pre trained, it was first thought sufficient to manually annotate and train the models with a 200 image dataset separated from the selected 2000 dataset.

The rest of which would then be validated by the model and given that the models in use also annotate or atleast mark the identified target to some extent with bounding boxes, the annotation would be by far automated and just had to be manually verified.

This would help in fine tuning and the bigger dataset would then be tested by the trained models.

However, the plan was changed due to the bad quality of the dataset affecting the accuracy of the model. And a freshly segregated top-view-images-only dataset without any bad and undesirable images would then be implemented without manual annotations, allowing to annotate the images automatically.

5.3 Model training and Fine-tuning

After all the steps of the pipeline were achieved, shall the models finally be trained with a smaller dataset, verified manually and fine tuned to a bigger set of images, finally testing it for the biggest dataset available for us. The models and the details of the training including evaluation of performance is to be found in the following dedicated section.

6 Model and Training

To evaluate different object detection approaches for ragwort detection, we experimented with two model families: YOLOv8 and a more recent DINOv3 + DEIMv2 pipeline. The models differ mainly in their architecture and in the way visual features are extracted and used for object detection.

6.1 DEIMv2 + DINOv3

As a more recent approach, we tested DEIMv2 combined with DINOv3. In this pipeline, DINOv3 is used as the visual feature extractor. It produces rich image representations (embeddings) that capture semantic and visual information about the input image. These representations are then used by the DEIMv2 detector to perform object detection.

This approach separates the feature extraction stage from the detection stage. Instead of learning all visual representations directly within the detection network, DEIMv2 can take advantage of the strong representations learned by DINOv3. This is particularly interesting for our task, where the model needs to

distinguish ragwort plants based on relatively subtle visual characteristics such as leaf structure, shape, and flower appearance.

We trained and evaluated the DEIMv2 + DINOv3 configuration on our annotated dataset and investigated its detection performance during training. The resulting model was also exported to a format suitable for integration with the application.

6.2 YOLOv8

As a second approach, we evaluated YOLOv8, a widely used one-stage object detection architecture. YOLOv8 performs feature extraction, object localization, and classification within a single detection pipeline, allowing it to perform inference efficiently.

We experimented with different YOLOv8 configurations and training parameters to determine how well the architecture performs on our ragwort dataset. The experiments included different image resolutions and data augmentation strategies. In particular, we investigated augmentations such as HSV transformations and random erasing, which encourage the model to rely less on specific colors and more on structural characteristics of the plant.

YOLOv8 was also used as a baseline for comparison with the DINOv3 + DEIMv2 approach. This allowed us to compare a well-established object detection architecture with a newer pipeline based on pretrained visual representations and the DEIMv2 detector.

6.3 YOLOv8 Fine-Tuning and Domain Adaptation

While the baseline YOLOv8s detector trained on a mixture of synthetic composites and general web imagery achieved competitive nominal metrics on synthetic benchmarks, initial field tests revealed severe domain discrepancies and shortcut learning. Specifically, the detector exhibited an excessive false-positive rate when encountering other yellow-flowering meadow flora and demonstrated fragile recall on vegetative rosettes where flowering heads were absent or occluded.

To overcome these failure modes, we executed a systematic, five-stage fine-tuning curriculum divided into two strategic phases:

1. **Phase I: In-Situ Domain Adaptation and Negative Regularization** (Stages 1–2) — targeting the elimination of false positives on non-target meadow flora while preserving high recall on real-world camera views.
2. **Phase II: Leaf Morphology and Color-Invariant Adaptation** (Stages 3–5) — training the network to detect *Jacobaea vulgaris* by leaf structure and rosette geometry rather than relying exclusively on yellow inflorescences.

6.3.1 Phase I: In-Situ Domain Adaptation and Precision Calibration

Motivation and The Yellow-Lookalike Dilemma. In agricultural practice, deploying a detector with high false-positive rates leads to erroneous actuator triggering (e.g., unnecessary herbicide micro-spraying or mechanical tillage on harmless or beneficial plants). When evaluated on a curated background verification set of 350 negative meadow images containing co-occurring lookalike flora (*data/facility*, featuring *Barbarea vulgaris*, *Ranunculus repens*, *Taraxacum officinale*, *Hypericum perforatum*, and *Brassica napus*), the baseline model generated 257 false positive bounding boxes across 179 distinct images. This translates to a **51.1% false-positive image rate** (179/350), rendering the baseline unviable for autonomous field operation.

Stage 1: Real-World Transfer with Hard Negative Background Regularization. In Stage 1, we fine-tuned the baseline checkpoint using 180 real field photographs captured from our target agricultural platform (*data/data_concatenated/Felix*) alongside 315 verified negative images from the facility dataset. Critically, each negative image was supplied with an empty (zero-byte) YOLO annotation file. In the YOLOv8 loss formulation, an image containing zero bounding boxes contributes exclusively to the background class penalty:

$$\mathcal{L}_{\text{det}} = \lambda_{\text{box}}\mathcal{L}_{\text{box}} + \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \lambda_{\text{dff}}\mathcal{L}_{\text{dff}} \quad (1)$$

When no ground-truth boxes are present, $\mathcal{L}_{\text{box}}$ and $\mathcal{L}_{\text{diff}}$ evaluate to zero, and the classification loss $\mathcal{L}_{\text{cls}}$ heavily penalizes any residual logits predicted over background textures and lookalike yellow petals. Training was conducted for 10 epochs using the AdamW optimizer with initial learning rate $\text{lr}_0 = 5 \times 10^{-4}$, cosine decay multiplier $\text{lrf} = 0.1$, and batch size 16.

As shown in Table 1, Stage 1 successfully suppressed false alarms: false positive boxes on the facility dataset dropped from 257 to 54 (an **79.0% reduction**), and the affected image rate fell from 51.1% to 12.9%. However, the large ratio of negative samples induced conservative classification behavior, resulting in lower recall on complex scenes (mAP@50 dropped from 0.2742 to 0.2242).

Stage 2: Weighted Resampling and Controlled Synthetic Undersampling. To resolve this sensitivity penalty and recover high detection recall without reintroducing false positives, Stage 2 introduced a structured data resampling policy:

- **Target In-Situ Oversampling (12 $\times$ multiplier):** The 180 verified real field images were oversampled twelve-fold (2,160 instances per epoch), accounting for **79.3%** of the total gradient update budget.
- **Synthetic Undersampling (9.2% share):** To prevent catastrophic forgetting of rare geometric orientations while avoiding synthetic-domain overfitting, synthetic images were strictly capped at 250 samples per epoch.
- **Background Regularization (11.5% share):** 315 facility negative images were retained at single weight (1 $\times$) to maintain continuous suppression of lookalikes.

Stage 2 trained for 10 epochs starting from the Stage 1 checkpoint with $\text{lr}_0 = 5 \times 10^{-4}$. As summarized in Table 1, Stage 2 successfully restored mAP@50 to 0.2631, increased the average confidence on real field images to its highest level (**0.865**), and reached a field plant detection rate of **99.5%** (199/200 images, 551 plants detected) while retaining an overall 54% false-positive reduction compared to the baseline.

Table 1: Quantitative comparison of Phase I fine-tuning stages evaluated at standard confidence ($\text{conf} = 0.25$) and under precision-calibrated operation ($\text{conf} = 0.35$).

Model Stage	mAP@50	mAP@50-95	Precision	Recall	Facility FP Rate	Field In-Situ Det.
<i>Evaluation at default threshold ($\text{conf} = 0.25$):</i>						
Baseline (Base)	0.2742	0.1730	0.8101	0.2791	51.1% (179/350)	99.0% (198/200, $\bar{c} = 0.839$)
Stage 1 (Real + Neg)	0.2242	0.1490	0.3296	0.1973	12.9% (45/350)	98.5% (197/200, $\bar{c} = 0.834$)
Stage 2 (Weighted)	0.2631	0.1664	0.3863	0.2145	26.6% (93/350)	99.5% (199/200, $\bar{c} = \mathbf{0.865}$)
<i>Operating point calibrated for high purity ($\text{Precision}_{\text{ragwort}} > 65\%$, $\text{conf} = 0.35$):</i>						
Baseline	0.2100	0.1340	0.6960	0.4910	42.3% (148/350)	99.0% (198/200, 527 boxes)
Stage 1	0.1248	0.0810	0.8280	0.2610	8.9% (31/350)	98.0% (196/200, 526 boxes)
Stage 2 (Optimal)	0.1901	0.1210	0.8130	0.4030	20.0% (70/350)	99.5% (199/200, 535 boxes)

Diagnostic Operating Point Calibration (Precision > 65%). Because industrial robotic weeders require strict bounds against collateral crop damage, we investigated the precision-recall trade-off across confidence thresholds $\text{conf} \in [0.25, 0.80]$. The baseline model failed the Precision > 65% requirement at its default operating point (Precision = 63.4% at $\text{conf} = 0.25$). Conversely, Stages 1 and 2 met this criterion directly from $\text{conf} = 0.25$ (80.7% and 77.3%, respectively). Calibrating all models to a unified operating point satisfying Precision > 65% ($\text{conf} = 0.35$) demonstrated that Stage 2 delivers the optimal Pareto trade-off: **81.3% precision** and **40.3% recall** on target plants, detecting 535 plants across 199/200 field scenes while halving false-positive alarms on non-target flora.

6.3.2 Phase II: Leaf Morphology and Color-Invariant Adaptation

The Need for Vegetative Feature Extraction. In European agricultural monitoring, *Jacobaea vulgaris* must be identified and removed during its early rosette and pre-flowering stages, as the plant develops severe

hepatotoxicity before inflorescences emerge. Furthermore, models relying strictly on flower color fail when blossoms are occluded by grasses or senesce late in the season. We therefore developed a secondary fine-tuning pipeline (`fine_tuning_leaves.py`) to compel the detector to utilize leaf lobation, margin serration, and rosette geometry.

Stage 3: High-Resolution Spatial Feature Recovery (imgsz = 1024). Standard downsampling to 640×640 pixels discards the high-frequency spatial cues of ragwort foliage, such as the distinctive pinnatifid lobes and ruffled leaf margins. In Stage 3, we increased the input resolution to 1024×1024 pixels, initializing weights from the Stage 2 checkpoint. Training was executed with AdamW ($\text{lr}_0 = 5 \times 10^{-4}$, $\text{lrf} = 0.1$, batch size 8). As evidenced in Table 2, evaluating the original Stage 2 model directly at 1024 resolution yielded only $\text{mAP@50} = 0.0628$ due to unadapted convolutional anchor scales. Stage 3 fine-tuning rapidly adapted the feature pyramids to the higher resolution, boosting mAP@50 to **0.1896** and elevating field plant detection to **100.0%** (623 detected plants across 200/200 images).

Stage 4: Structural Occlusion and Color Jitter Augmentation. To explicitly penalize reliance on flower heads, Stage 4 integrated structural and photometric augmentations starting from the Stage 3 checkpoint:

- **Random Erasing / Cutout (erasing = 0.40):** Rectangular patches masking up to 40% of image regions, frequently occluding flowers and forcing backpropagation through foliage features.
- **Photometric Distortions (hsv_s = 0.8, hsv_v = 0.5):** Extreme saturation and value jitter simulating varying diurnal lighting conditions and neutralizing bright yellow contrast.

Trained with $\text{lr}_0 = 3 \times 10^{-4}$ at $\text{imgsz} = 1024$, Stage 4 elevated average detection confidence on real field images to its highest recorded value ($\bar{c} = \mathbf{0.751}$), confirming enhanced feature robustness under partial occlusion.

Table 2: Progression of Phase II leaf-focused fine-tuning evaluated at high resolution (1024×1024) on the unified validation set and in-situ field images.

Fine-Tuning Stage	Resolution	mAP@50	Precision	Recall	Field Plant Count	Avg. Confidence
Stage 2 (Zero-shot 1024)	1024×1024	0.0628	0.1382	0.0879	529 plants (97.0%)	0.689
Stage 3 (High-Res Scale)	1024×1024	0.1896	0.2495	0.2187	623 plants (100.0%)	0.705
Stage 4 (Morphology Aug.)	1024×1024	0.1870	0.2552	0.2093	623 plants (100.0%)	0.751
Stage 5 (No-Yellow Color)	1024×1024	0.1755	0.2713	0.1848	664 plants (100.0%)	0.727

Stage 5: Non-Invasive Color Invariance via HSV Flower Desaturation. To definitively decouple the model from yellow chromatic cues, Stage 5 engineered a non-invasive synthetic domain transformation without altering ground-truth bounding box coordinates. A non-destructive copy of the training dataset was created where a selective transformation in HSV color space was applied:

$$S'(x, y) = \begin{cases} 0, & \text{if } H(x, y) \in [18^\circ, 36^\circ] \text{ and } S(x, y) > 0.25 \\ S(x, y), & \text{otherwise} \end{cases} \quad (2)$$

Pixels falling within the yellow petal spectrum ($H \in [18, 36]$) were mapped to neutral achromatic grays ($S \rightarrow 0$), simulating post-flowering or rosette conditions. In contrast, 100% of vegetative green foliage ($H \in [38, 85]$), stems, and background soil were perfectly preserved.

Continuing from Stage 4 weights for 8 epochs, the resulting Stage 5 model achieved **100.0% detection coverage** on the target field set (664 plants detected across 200/200 images, $\bar{c} = 0.727$). Even when yellow flower cues were entirely absent, the detector attained a class-specific ragwort precision of **55.1%** and mAP@50 of **35.3%** purely from leaf contour, lobe geometry, and rosette habit. This confirms that the fine-tuned YOLOv8 network operates as a true morphological detector rather than a simplistic yellow-flower spotter.

7 Actual outcome that can be achieved

The result of this work is a model able to detect *Jacobaea vulgaris* both with and without flowers, at different growth stages and from different viewing angles. Together with the model we built a complete and repeatable process that turns an open biodiversity archive into a dataset ready for training, and two detector setups that run on that dataset.

All plant images come from the Global Biodiversity Information Facility (GBIF), a public database that collects records of *Jacobaea vulgaris* with their geographic coordinates. The records are supplied by citizen-science platforms and by national observation networks. From the GBIF media export we took a random sample of 10 000 records. After removing empty entries and duplicate links, 9946 unique image addresses remained, and 9944 of them were downloaded successfully. Every GBIF record states its licence and its rights holder, so this information stays with the images through the whole process. This is what makes the resulting dataset fit for publication, and not only for internal use.

The following steps were implemented and run from start to finish:

- Each candidate image is passed through the DINOv3 vision model, which describes it with 384 numbers. Such a description captures the content of an image far better than simple colour statistics, and it needs no labels.
- A reference set of 223 photographs taken from above defines what a useful image looks like. Each candidate is scored by how similar it is to this set, which reduced the pool to 3 000 images that match the intended camera view.
- Inside this smaller pool, a facility location problem solved with a greedy k -center algorithm picks the images that best cover the variety of the data. For 200 images the selection reached an average coverage of 0.92 and a worst-case coverage of 0.84. The order of the selection does not change when a different number of images is requested, so the result also works as a priority list for annotation.
- Annotated plants are reused to build synthetic training images. A plant is pasted onto a verified grassland background and the bounding box is taken from the pasted area, so the label is exact and costs no annotation work.
- The dataset is exported in both YOLO and COCO format. This feeds a YOLOv8 baseline and a DEIMv2 detector that uses the same DINOv3 model as the selection step, so a fast convolutional detector and a transformer detector can be compared on exactly the same data.

The main benefit of this result is practical. The limit of the project is manual annotation, not computing power. The process turns about ten thousand unlabelled records into a few hundred images that deserve a person's attention, and it multiplies every annotated plant through synthesis.

8 How much of the goal could be achieved

The goal was the automatic detection of ragwort in field images taken from above. The result meets this goal in part, and it is worth stating clearly which part.

The data problem is solved well enough that training is no longer blocked. A selected dataset with tracked licences is ready for annotation, the selection procedure can be repeated and adjusted through its parameters, and both detectors train and are evaluated on the same exported data. The evaluation reports mean average precision, but also precision, recall and the false positives and false negatives for each image, so mistakes can be examined one by one instead of being hidden in an average.

The dataset still carries the bias of its source. GBIF records are photographs taken by people, and people photograph plants when they are easy to see, which for ragwort means the flowering period. In our own selection the typical image devotes about one fifth of its area to yellow flowers, and fewer than one image in ten shows no flower at all. Coverage of the non-flowering stage therefore depends on deliberate choices: the relevance filter uses image similarity rather than colour, and the reference set has to include rosettes.

[To be added: measured results.]

9 What can be done in the future

The clearest next step is to build our own image database instead of relying only on public sources. Every dataset used in this work was collected by other people for other purposes, and this is the origin of every limitation described above: the viewing angle, the distance to the plant, the lighting and the share of flowering plants were all decided by whoever took the photograph. Producing our own images removes this dependency at the source. We did not have enough time for such a collection during this stage of the project, but it is the natural continuation of the work.

Such a database should be recorded with the same camera and at the same mounting height that the final system will use, so that the training images show exactly what the device will see. It should contain plants both with and without flowers, which in practice means several field sessions spread across the season: rosettes appear well before the flowering period, and a single visit in summer cannot capture them. Recording in different weather and at different times of day is equally important, because bright sunlight, overcast sky and wet vegetation change the appearance of the sward far more than the plant itself. Every image should keep its date and location, so that the material can later be divided by season or by field rather than at random.

The second part of this task is a validation set recorded in the same way. At present the project has no test data that matches the deployment conditions, which means that no reported number can be fully trusted. Even a few hundred frames taken with the real camera would be enough to measure how well the model transfers from web photographs to the field, instead of assuming that it does. This set should stay separate from the training material and should contain only real images, so that it remains a fair measure of progress.

The existing pipeline supports this extension without modification. The selection, annotation and weighting steps operate on any folder of images, so our own recordings can be added as a new source, and the selection procedure can be used again to decide which of the newly collected frames are worth annotating.

10 Resources supporting our approach

The choices described above are not arbitrary. Each of them follows established work. GBIF is the standard source of species occurrence data and provides both the images and the licence information needed to share them. Using a self-supervised model as a general image descriptor follows DINOv3, whose features are made to work without fine-tuning, which is exactly what an unlabelled pool of images requires. Choosing a subset that covers a feature space is the classical k -center problem, for which the greedy algorithm has a known quality guarantee, and its use for deciding what to annotate was established by the core-set approach to active learning. Building training images by pasting objects onto new backgrounds is a known augmentation method with reported improvements in detection and segmentation. On the detector side, YOLOv8 is a widely used real-time baseline, while DEIM and its successor DEIMv2 represent the transformer alternative, the latter combining the detection head with the same DINOv3 model. Finally, a separately published drone dataset of *Jacobaea vulgaris* and meadow images, recorded from about 11 metres with a ground resolution of 3.18 millimetres per pixel, provides verified negative examples in exactly the view we aim for.